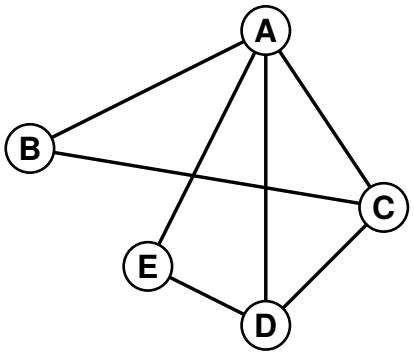
A network is dots joined by links. You may move the dots and bend the links, but each link stays joined to its own two dots and never lies across a dot. A crossing is a place where one link lies across another, and no more than two links may cross at the same place. A network is untangled when it has no crossings.

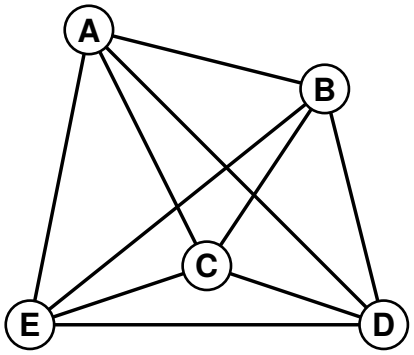
Problem 1: Make each network with dots and pipe cleaners, or draw it, and try to untangle it. In the box beside it, draw it untangled or write “stuck”.

1



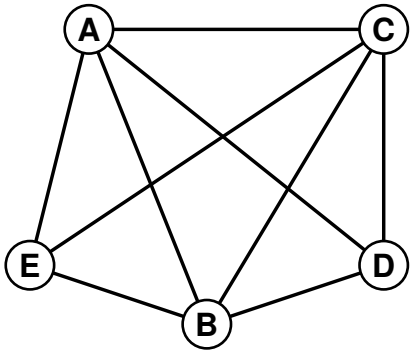
A – B B – C
A – C C – D
A – D D – E
A – E

2



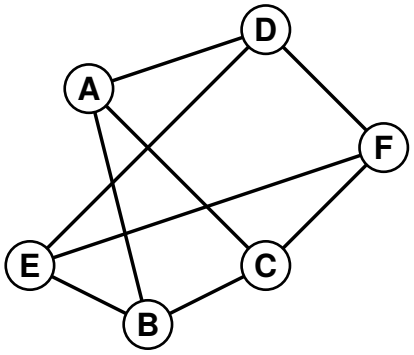
A – B B – D
A – C B – E
A – D C – D
A – E C – E
B – C D – E

3



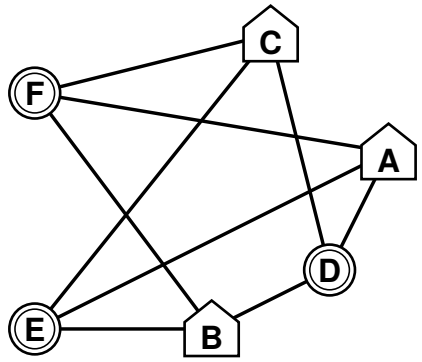
A – B B – D
A – C B – E
A – D C – D
A – E C – E
B – C

4



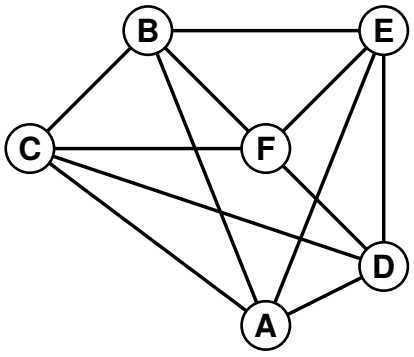
A – B C – F
A – C D – E
A – D D – F
B – C E – F
B – E

5



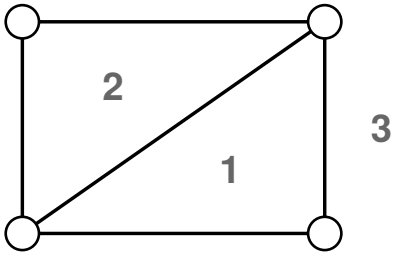
A – D B – F
A – E C – D
A – F C – E
B – D C – F
B – E

6



A – B B – F
A – C C – D
A – D C – F
A – E D – E
B – C D – F
B – E E – F

Problem 2: A region is a part of the paper fenced in by links. The outside counts as a region too. Count the dots, links and regions in each network you untangled. Find a rule connecting the three numbers that works for every untangled network in one piece.

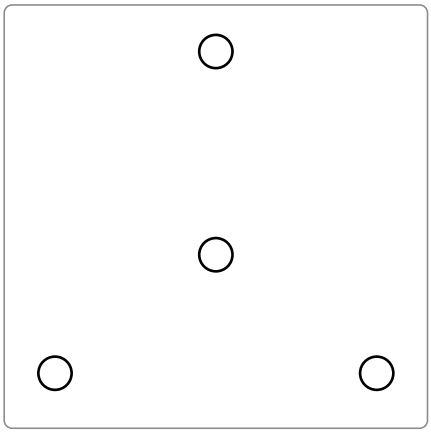
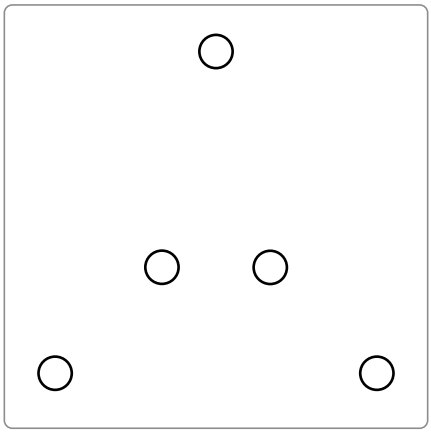
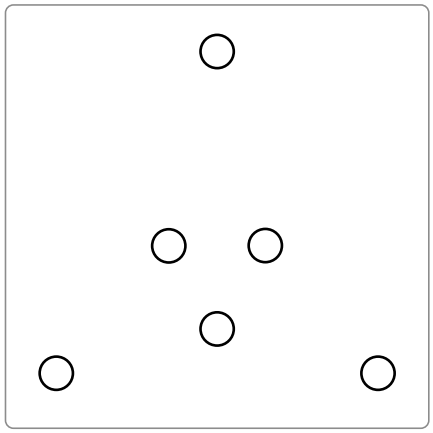
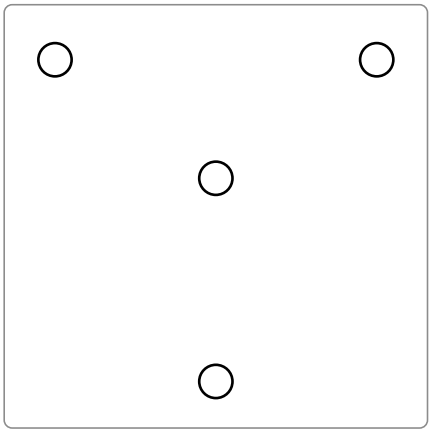
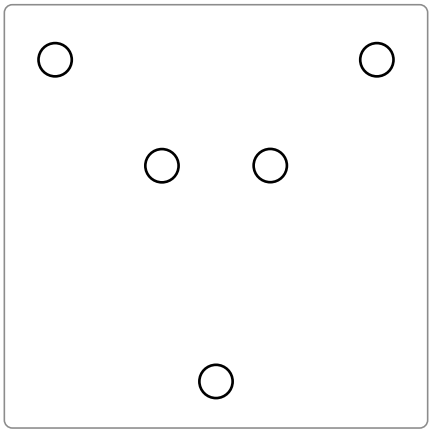
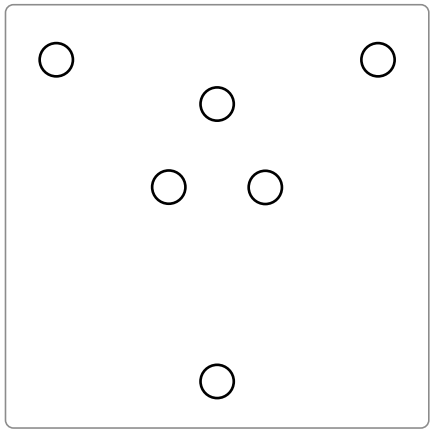


4 dots, 5 links, 3 regions

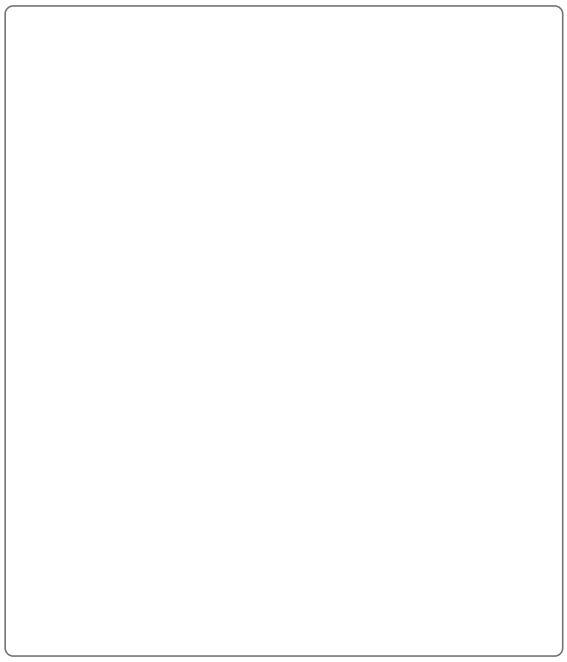
Network	Dots	Links	Regions

Problem 3: Two players take turns drawing a link between two of the dots below. A link may not make a crossing, and two dots may not have two links between them. The player who cannot draw a link loses. Play with 4 dots, with 5 dots and with 6 dots. For each number of dots, which player wins?

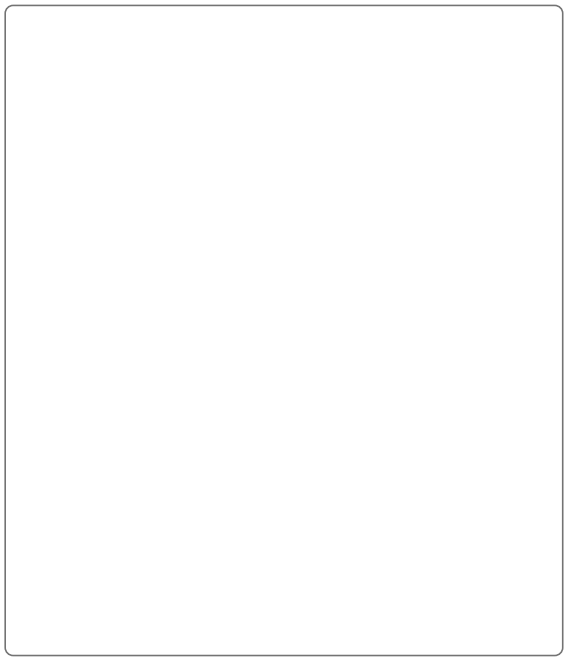
	First player wins	Second player wins
4 dots		
5 dots		
6 dots		

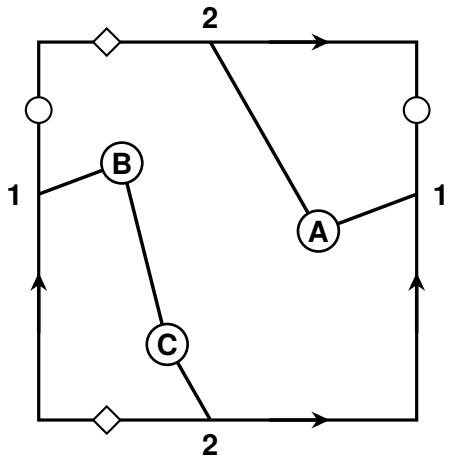
Problem 4: Explain why network 2 can never be untangled.

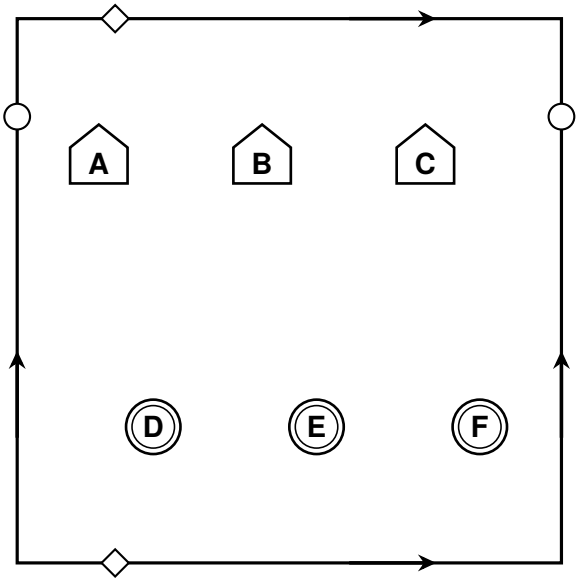


Problem 5: Explain why network 5 can never be untangled.

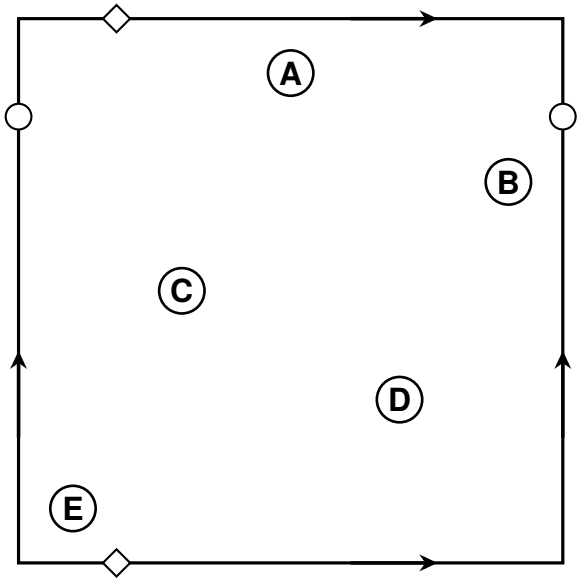


Problem 6: On a portal board, a link that goes out through one side comes back in at the same place on the opposite side. Draw each of these networks on a portal board with no crossings.

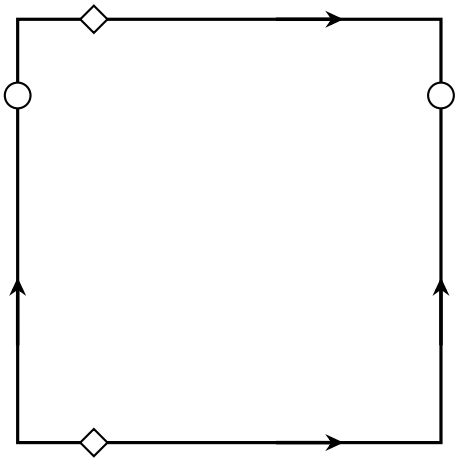




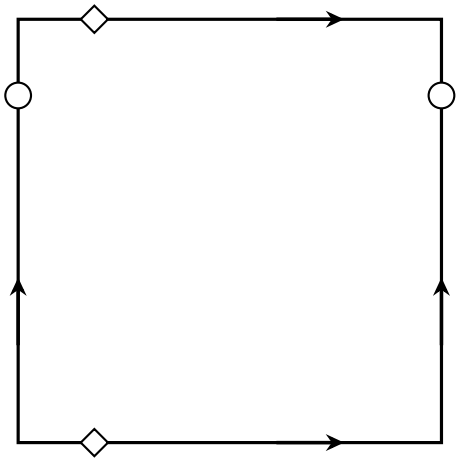
Network 5



Network 2

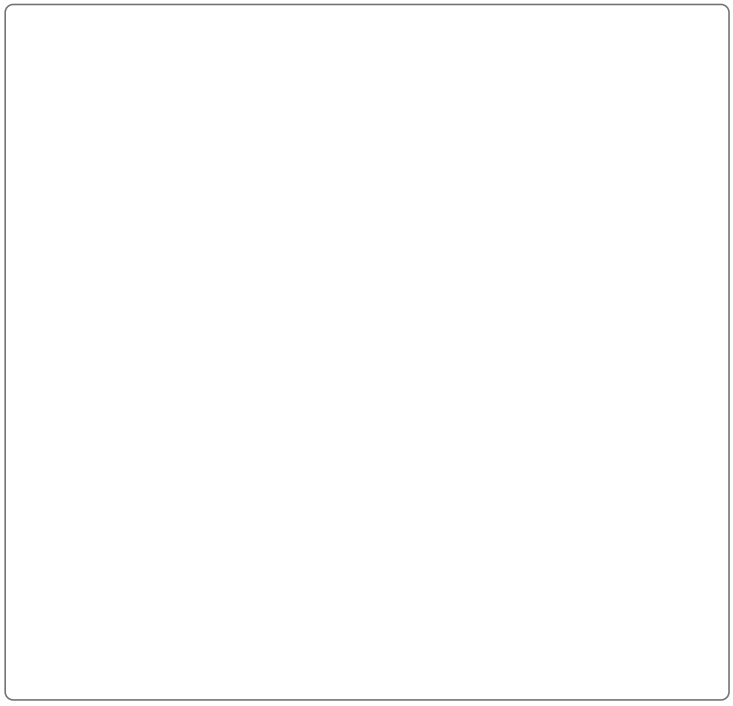


Extra board



Extra board

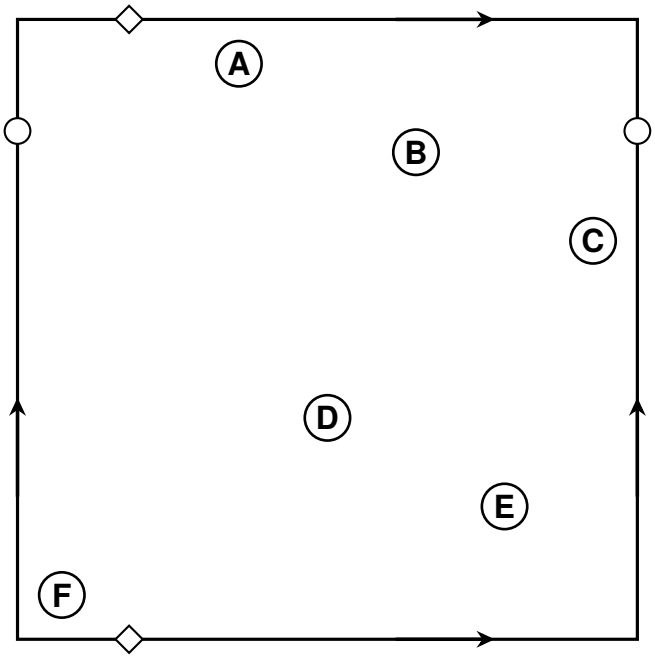
Problem 7: Draw the network with 6 dots, each joined to every other dot, with as few crossings as you can. Explain why nobody can do better.

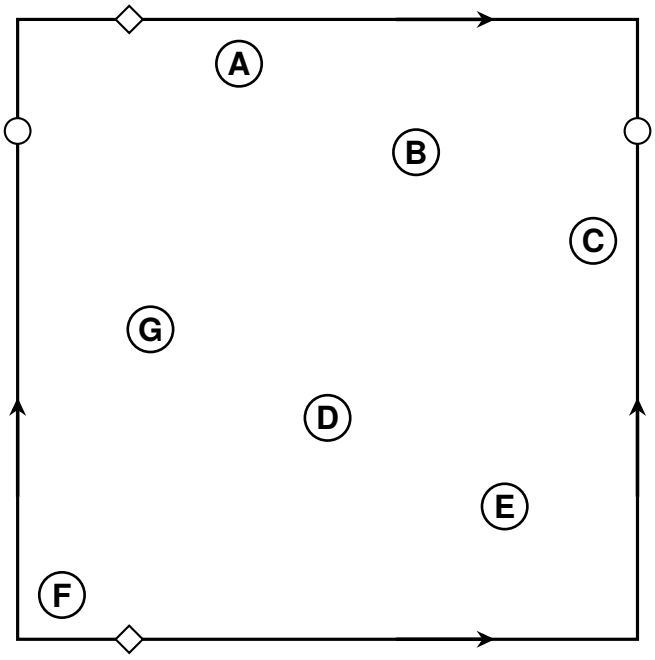


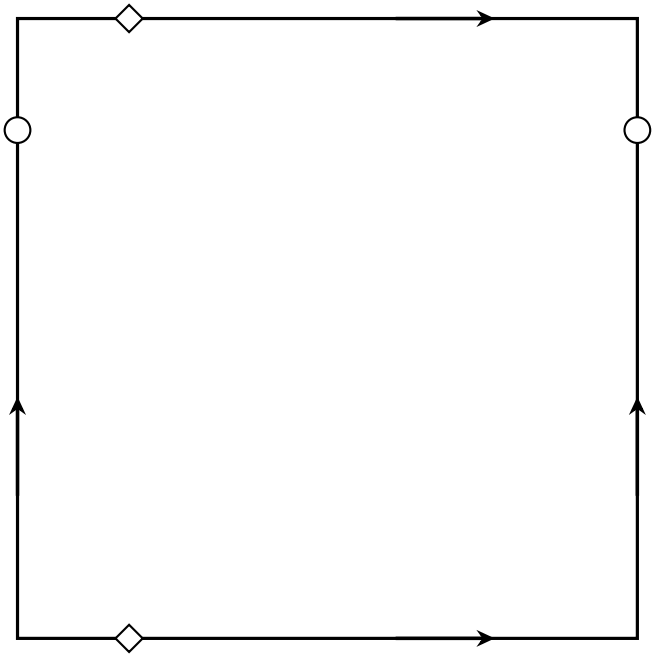
Problem 8: Which player wins the game in Problem 3 when it is played with 10 dots, and which player wins with 11 dots? Explain how you can tell before the game starts.



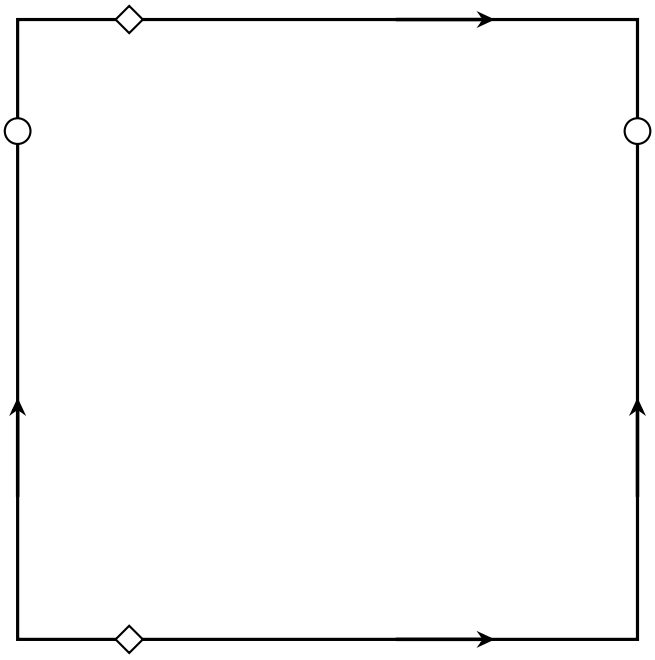
Problem 9: Join each of these 6 dots to every other dot on a portal board with no crossings. Then do the same with the 7 dots.







Extra board



Extra board